

1.3

1. When a coin is tossed 3 times. the sample space would be $2^3 = 8$

$$S = \{HHH, HHT, HTH, TTT, THT, TTH, HTT, THH\}$$

2. 3 out of 8 have two heads
HHT, THH, HTH

3. $A = \{HHT, THH, HTH\}$

$$B = \{TTH, THT, HTT\}$$

$$A \cup B = \{HHT, THH, HTH, TTH, THT, HTT\}$$

$$A \cap B = \emptyset = \text{Empty or Null set}$$

1.4

Soln:- Here

$$P(E) = 0.55$$

$$P(D) = 0.45$$

$$P(E \cup D) = 0.7$$

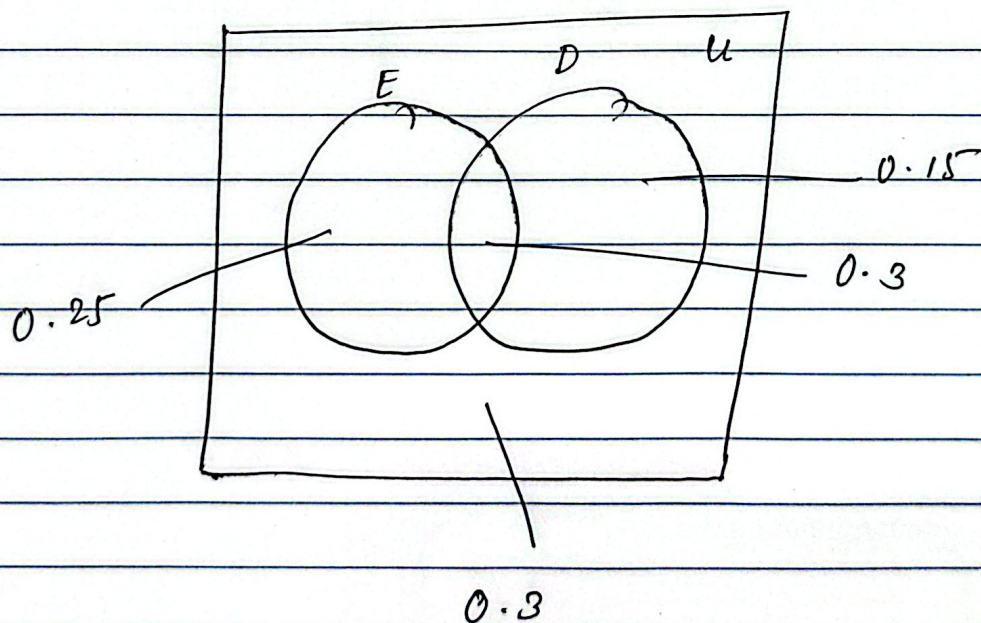
1. $P(E \cap D) = ?$

$$P(E \cup D) = P(E) + P(D) - P(E \cap D)$$

$$0.7 = 0.55 + 0.45 - P(E \cap D)$$

$$P(E \cap D) = 1 - 0.7 = 0.3 = 30\%$$

2. $P(\overline{E \cup D}) = 1 - P(E \cup D) = 1 - 0.7 = 0.3$



Adding $(0.25 + 0.15 + 0.3 + 0.3)$ gives 1.